

# Probability and Random Processes

## Sequences of random variables.

### Exercises

1. (a) Demonstrate, using the continuity theorem, that the probability distribution of a random variable  $X_n \sim \text{Bin}(n, \lambda/n)$  converges to a  $\text{Po}(\lambda)$  distribution as  $n \rightarrow \infty$ .  
(b) Given a book with 300 pages containing a total of 200 typographical errors, apply the Poisson approximation to determine the probability that a specific page contains more than one error.
2. Consider the sequence  $\{X_n: n \geq 1\}$  of normal random variables such that  $\mathbb{E}(X_n) = 1/n$  and  $\text{Var}(X_n) = 1$ . Demonstrate that  $X_n \xrightarrow{d} N(0, 1)$  as  $n \rightarrow \infty$ . Additionally, if  $f_n(x)$  denotes the probability density function of  $X_n$ , determine  $\lim_{n \rightarrow \infty} f_n(x)$ .
3. Let  $\{X_n: n \geq 1\}$  be random variables that are uniformly distributed on  $[0, 1/n]$ , and let  $X$  be a random variable such that  $\mathbb{P}(X = 0) = 1$ . Prove that  $X_n \xrightarrow{d} X$  as  $n \rightarrow \infty$ .
4. Let  $X$  be a random variable uniformly distributed on  $[0, 1]$ , and for each  $n \geq 1$ , let  $X_n$  denote a random variable uniformly distributed on  $[0, 1 + 1/n]$ . Demonstrate that  $X_n \xrightarrow{d} X$  as  $n \rightarrow \infty$ .
5. Determine the limiting distribution of the sequence  $\{X_n: n \geq 1\}$ , where each  $X_n$  is uniformly distributed on  $(-1/n, 1/n)$ . Additionally, demonstrate that this sequence converges in probability and almost surely.
6. Let  $X_1, X_2, \dots$  be a sequence of independent random variables, each with density function  $f_{X_n}(x) = x/2$  for  $0 < x < 2$ . Define  $M_n = \max(X_1, \dots, X_n)$ . Determine the limiting distribution of the sequence  $M_1, M_2, \dots$ . Additionally, assess whether this sequence converges in probability and in mean square.
7. Let  $S_n = \min(X_1, X_2, \dots, X_n)$ , where  $\{X_k: k \geq 1\}$  is a sequence of independent random variables, each one uniform on the interval  $(0, 1)$ . Prove that the sequence  $\{S_n: n \geq 1\}$  converges to 0 in mean square and in probability.  
*Hint:*  $\int_0^1 t^m (1-t)^n dt = (m! n!)/(m+n+1)!$
8. Let  $\{X_k: k \geq 1\}$  be a sequence of independent random variables, uniformly distributed on  $(0, 1)$ , and let  $Z_n = \max(X_1, X_2, \dots, X_n)$ .  
(a) Prove that  $\mathbb{P}(Z_n \leq z) = z^n$ ,  $0 < z < 1$ .  
(b) Let  $U_n = n(1 - Z_n)$ . Prove that the distribution function of  $U_n$  converges to the  $\text{Exp}(1)$  distribution as  $n \rightarrow \infty$ .  
*Hint:*  $\lim_{n \rightarrow \infty} (1 + a/n)^n = e^a$ .
9. (*Empirical distribution functions*) Consider  $X_1, X_2, \dots, X_n$  as  $n$  independent random samples drawn from a random variable  $X$  with probability distribution function  $F_X(x)$ . For each  $x \in \mathbb{R}$ , define  $F_n(x) = N_n(x)/n$ , where  $N_n(x)$  denotes the number of samples with values less than or equal to  $x$ .  
(a) Demonstrate that  $F_n(x) \xrightarrow{\mathbb{P}} F_X(x)$ .  
(b) Show that  $\sqrt{n}(F_n(x) - F_X(x))$  converges in distribution to a normal distribution. Specify the mean and variance of the limiting distribution.
10. Let  $X_1, \dots, X_n$  be random samples of a random variable  $X$  uniformly distributed on  $(0, \theta)$ . Let  $T_n = 2\bar{X}_n$ ,  $V_n = \sqrt{n}(T_n - \theta)$ , and  $W_n = \max(X_1, \dots, X_n)$ . Prove that  
(a)  $T_n \xrightarrow{\mathbb{P}} \theta$ . (Thus,  $T_n$  is a consistent estimator of  $\theta$ .)

(b)  $V_n \xrightarrow{d} N(0, \theta^2/3)$ , where the parameter  $\theta^2/3$  denotes the variance.

(c)  $W_n \xrightarrow{\mathbb{P}} \theta$ .

11. Let  $Z_1, Z_2, \dots, Z_k, \dots$  denote a sequence of independent  $N(0, 1)$  random variables. Define  $V_n = \sum_{k=1}^n Z_k^2$ . The variable  $V_n$  follows a chi-square distribution with  $n$  degrees of freedom, and its characteristic function is  $M_{V_n}(\omega) = (1 - 2i\omega)^{-n/2}$ .

(a) Find the limit in probability and in mean square of the sequence  $\{2V_n/n : n \geq 1\}$ .

(b) Now let

$$T_n = \frac{Z_n}{\sqrt{V_n/n}}.$$

Find the limit in distribution of the sequence  $\{T_n : n \geq 1\}$ .

12. Let  $\{X_n : n \geq 1\}$  be a sequence of independent and identically distributed random variables with distribution  $N(0, \sigma^2)$ . Consider

$$Y_1 = \frac{X_1}{2}, \quad Y_2 = \frac{X_2 + Y_1}{2}, \quad \dots, \quad Y_n = \frac{X_n + Y_{n-1}}{2}, \quad \dots$$

(a) Prove that the characteristic function of  $Y_n$ ,  $n \geq 1$ , is

$$M_n(\omega) = \exp \left( -\frac{\omega^2 \sigma^2}{2} \left( \frac{1}{4} + \dots + \frac{1}{4^n} \right) \right).$$

(b) Find the limit in distribution of the sequence  $\{Y_n : n \geq 1\}$ .

13. Let  $X_n \sim \text{Po}(n)$ ,  $n \geq 1$ . Prove that

$$\frac{X_n - n}{\sqrt{n}} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty,$$

(a) using the Central Limit Theorem;

(b) using characteristic functions. [*Hint:*  $e^z = 1 + z + z^2/2 + o(|z|^2)$ ,  $z \in \mathbb{C}$ .]

14. Let  $X_1, \dots, X_n$  be independent random samples of a random variable  $X$  that is  $\text{Po}(\lambda)$ -distributed. Let  $\theta = \mathbb{P}(X = 0) = e^{-\lambda}$ . Prove that, as  $n \rightarrow \infty$ ,

(a)  $e^{-\bar{X}_n} \xrightarrow{\mathbb{P}} \theta$ .

(b)  $\sqrt{n} (e^{-\bar{X}_n} - \theta) \xrightarrow{d} N(0, e^{-\lambda} \sqrt{\lambda})$ , where  $e^{-\lambda} \sqrt{\lambda}$  denotes the standard deviation.

15. Let  $\{X_n : n \geq 1\}$  be a sequence of discrete random variables such that  $\mathbb{P}(X_n = k) = (1 - p_n)^k p_n$ , where  $p_n = 2/(n + 2)$  and  $k = 0, 1, 2, \dots$

(a) Show that the characteristic function of  $X_n$  is

$$\frac{2}{2 + n(1 - e^{i\omega})},$$

and deduce that  $\{X_n : n \geq 1\}$  does not converge in distribution.

(b) Prove that

$$\frac{X_n}{n} \xrightarrow{d} \text{Exp}(\lambda).$$

Which is the value of the parameter  $\lambda$ ? [*Hint:*  $e^{it} = 1 + it + o(t)$ .]

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## Answers

1. 0.1443
  2.  $\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$
  - 3.
  - 4.
  5. 0
  6. 2, yes, yes
  - 7.
  - 8.
  9. 0,  $F_X(x)(1 - F_X(x))$
  - 10.
  11. (a) 2    (b)  $N(0, 1)$
  12. (b)  $N(0, \sigma^2/3)$
  - 13.
  - 14.
  15. (b)  $\lambda = 2$
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